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Task Description:	Understanding of qualitative phenomena based on experiments supported by accurate computational simulations (steady and unsteady) for quasi-steady annular/stratified internal condensing flows in micro-gravity and terrestrial environments is essential for good design and performance of condensers in applications such as Rankine-cycle based power (nuclear) plants, Thermal Management systems, Looped Heat Pipes, Capillary Pumped Loops, etc. Condenser-sections in these devices often achieve complete or incomplete condensation by the exit under unspecified or specified (as being proposed) exit conditions while the inlet (pressure and flow rate) and wall conditions are typically always known or specified. These condensing flows' attainability/stability of steady flows, intrinsic waviness, noise-sensitivity (primarily due to minuscule transverse vibrations of the condensing surface), and exit-condition sensitivity critically depend on the presence or absence of gravity in driving the condensate. Zero-gravity simulations establish that the parameter zone within which steady flows exist for any space-based condenser, experiencing full or partial condensation with unspecified (free) exit condition (as this is a typical exit condition for condensers in many thermal systems), is very small. Thus achieving steady flows in such space-based condenser employments is very difficult (if not next to impossible) unless new technology – to be suggested by the current and planned research on exit-condition based flow categorizations - is employed. The ground based experiments and supporting simulations (both for terrestrial and microgravity environments) are to explore how exit-condition effects can lead to different steady and unsteady/oscillatory flows. The unsteady/oscillatory flows may arise from changes in flow regime within the condenser or from system-instabilities (due to interaction of condenser with other upstream or downstream components of the system). Although the proposed research involves fundamental investigations of flows inside relatively simpler tube and channel geometries, the qualitative understandings/results and tools that will be obtained are going to be sufficiently fundamental and general in nature that they will provide an important framework for facilitating the design, incorporation, and operation of new realistic space-based or ground-based condensers (that is condensers that employ more complex but realistic geometries such as SWEP/Tranter plate heat-exchangers, shell-and-tube type, etc.) in planned systems. Task Objectives Experimental • Complete experimental investigation of attainability/stability and exit-condition issues for downward and steady internal condensing flows inside a vertical tube. Following this gravity driven condensate flow investigations, initiate similar experimental investigations of gravity-to-shear driven internal condensing flows in inclined-to-horizontal channel (i.e. rectangular cross-section of suitable aspect-ratio) geometries which experience condensation on its bottom wall. • Ensure that the ground based experimental facility is well instrumented and progressively capable of providing more and more detailed information that can be synthesized with results obtained from the computational capability being developed for these flows. • Complete ongoing invention efforts and develop a non-intrusive film thickness sensor (based on principles of fluorescence and fiber-optics) that is capable of measuring time-varying thickness of the dynamic condensate. Verify such an invention by independent thickness measurements. Make progress on employing such sensors for the gravity driven annular flow cases in a vertical tube geometry and/or inclined-to-horizontal channel (i.e. rectangular cross-section of suitable aspect-ratio) geometry that experiences condensation on its bottom wall. Computational • Complete development of accurate simulation tool for laminar vapor/laminar condensate internal condensing flows for channel or vertical tube geometries. • Ensure that the simulation tool accurately locates the interface (under smooth or wavy conditions) for terrestrial and zero-gravity environments and meets various convergence and grid independence criteria for both specified and unspecified exit condition cases. • Ensure that the simulation tool is capable of reliably reproducing known results for classical benchmark problems (such as Nusselt problem of film condensation on a vertical plate, forced vapor flow experiencing film condensation on a horizontal plate, etc.) and, also, capable of providing new non-linear stability results consistent with known experimental results for these classical problems. • The verifications of the two dimensional code on classical problems as well as through successful comparisons of the simulation results for the ongoing ground based experiments are necessary for further enhancements (compressibility effects under specified exit conditions, three dimensional simulations, etc.) of the code. The verifications will also make simulation tool based zero gravity predictions more trustworthy. • Eventually the simulation tool should be reliable enough to help plan proper space based experiments and, also, to help reduce the number of design iterations needed to have a condenser operate successfully in a zero gravity environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Reliable existing zero-gravity simulations indicate that the parameter zone within which steady flows exist for any space-based condenser that experiences full or partial condensation with an unspecified (free) exit condition (as this is a typical exit condition for condensers in many thermal systems) is often non-existent to small. Thus achieving steady flows in any space-based condenser employments is very difficult (if not next to impossible) unless new technology – to be suggested by the current and planned research on specified exit-condition scenarios - is employed. Earth Benefits The exit condition based flow categorizations and experimental results are also going to be useful for many new terrestrial systems (such as two-phase electronic or chip cooling) that need to employ a condenser with a steady and stable flow for the thermal system. While it is true that many terrestrial systems can be made to function by sufficient number of trial and error experiments, the ground based results from this research will help significantly in innovation

and improvement of superior new technologies for such systems. For example, a specified exit condition mode of operation is expected to allow steady and stable operation of many terrestrial condensers in novel systems (electronic cooling, etc.) while concurrently cutting down on the system's start up times.

Also, the proposed development of film thickness sensor will be valuable in the measurement processes that are important to the development of many terrestrial applications that involve film cooling, film evaporation, film flows in chemical process industries, etc.

Task Progress:

Condensing Flow Experiments
Reported experimental and computational results ([1]-[7]) confirm that both the flow features and heat transfer rates inside a condenser depend on the specification of inlet, wall, and exit conditions. The results show that the commonly occuring condensing flows' special sensitivity to changes in exit conditions (due to changes in exit pressure) arise from the ease with which these changes alter the vapor flow field in the interior. When an exit condition is changed from one steady value to another, the changes required for the interior vapor flow to attain a new steady flow are such that they do not demand removal of the new exit pressure imposition back to the original steady value - as is the case for incompressible single phase flows' original and "required" steady exit pressure. Instead, new steady flows are achieved through appropriate changes in the vapor/liquid interfacial configurations and associated changes in interfacial mass, heat transfer rates (both local and overall), and other flow variables. This only happens for the commonly occurring cases in which the condensing surface temperature (not heat flux) and inlet conditions remain approximately the same while the exit condition changes. In the context of flows of a pure vapor that experience film condensation on the inside walls of a vertical tube, the reported experimental and computational results provide important quantitative and qualitative understanding as well as exit-condition based categorization of the flows.

Experimental results ([5]) and selected relevant computational results that have been reported reinforce the fact that there exist multiple steady solutions (with different heat transfer rates) for multiple steady prescriptions of the exit-condition – even though the other boundary conditions do not change. However, for some experimental situations that do not fix any specific value for the exit condition (say exit pressure) but allow the flow the freedom to choose any exit pressure value within a certain range, experiments confirm the computational results that, given enough time, there typically exists, under normal gravity conditions, a self selected "natural" steady flow with a "natural" exit condition. This happens if the vapor flow is seeking (or attracted to) a specific exit condition and the conditions downstream of the condenser allow the vapor flow a range of exit conditions that includes the specific "natural" exit condition of choice. However, for some unspecified exit condition cases involving partial condensation, even if computations predict that a "natural" exit condition choice exists, the experimental arrangement employed in this research does not allow the flow to approach its steady "natural" exit condition value and, instead, it only allows oscillatory exit conditions leading to an oscillatory flow. For the reported experiments, these oscillatory pressures are induced and imposed by instabilities in the system components downstream of the condenser. This case serves as an example of how system instabilities may affect the performance of condensers.

Quite a few of the reported experimental results ([5]), both qualitative and quantitative, are supported by accurate, numerical solutions of the complete and exact set of two-dimensional governing equations for steady and unsteady laminar-vapor/laminar-condensate internal condensing flows of pure vapor (FC-72 and R-113) inside a vertical tube and a channel. The film condensation is on the inside wall of a tube or one of the walls of a channel (the lower wall in the case of a downward sloping channel in normal gravity). Computations ([1]-[4], [6]) find that exit condition specification is very important. The simulations are also able to characterize the flows' sensitivity or insensitivity to the exit condition (exit-condition sensitive cases depend upon the flow downstream of the condenser). If well defined natural steady/quasi-steady flows exist – as is shown to be the case for gravity dominated or strong shear dominated condensate flows that exhibit the presence of an attractor, or long term parabolicity, up to the exit location – the computations are able to predict both the natural exit condition and any point of transition (from stable to unstable or smooth to wavy behavior) that may exist within this zone. Results on the role of surface tension and sensitivity to ever-present minuscule noise of the condensing surface have also been reported ([1]-[4]). The above results lead to the following (see [5]) important, exit-condition based, flow categorizations that must necessarily supplement other existing categorizations as to whether the flows are shear dominated or gravity dominated, internal or external, smooth or wavy at the interface, laminar or turbulent in the two phases, etc.

Category I (with complete or incomplete condensation under specified exit conditions)

• Prescribed or known values of inlet mass flow rate Min , inlet pressure pin, and inlet temperature (at saturation temperature Tsat(pin) or at some superheat) at all times t. This means the prescription could be steady or unsteady.

• Prescribed or known wall temperatures Tw(x,t) < Tsat(pin) for all x-locations over which film condensation occurs. Typical wall temperature conditions of interest are steady, but unsteady conditions are relevant to start-up and shutdown.

• Prescribed or known exit condition. For example, for steady exit conditions, exit pressure pe = constant, which is equivalent, for partial condensation cases, to setting exit mass quality Ze = Mv/Min = an appropriate constant, where Mv is the vapor mass flow rate at the exit at time t.

Category II (with complete or incomplete condensation under unspecified exit conditions)

• Prescribed or known values of inlet mass flow rate , inlet pressure pin, and inlet temperature (at saturation temperature Tsat(pin) or at some superheat) at all times t.

• Prescribed or known wall temperatures Tw(x,t) < Tsat(pin) for all x-locations over which film condensation occurs.

• No exit condition is prescribed, except for some system hardware limitations that may restrict the range of exit conditions available to the condenser.

Category III (complete condensation involving special specified conditions at the inlet and the exit). Though this is a special form of Category I flows, it is listed separately because it employs a different experimental set-up and has been extensively investigated in the literature for oscillatory condensing flows.

• In this case there is a constant pressure reservoir, with a high pressure, pTank-in, that feeds the vapor flow (at inlet pressure pin, temperature Tv-in, and density rho-v-in) into the test section through an inlet valve (with valve coefficient

ki). This requires that the inlet pressure, pin, satisfy: pin = pTank-in - ki*Min^2/rho-v-in

• Also there is a constant pressure exit tank with a lower pressure, pTank-exit, to which the condensate flows through an exit valve of valve coefficient ke. The exit valve handles an all liquid flow because this case is only for complete condensation flows. At the exit of the condenser, the flow rate is Mexit at any time, t, and the liquid density is rho-L-exit. This requires that the test section exit pressure, pexit, satisfy: pexit = pTank-exit + ke*Mexit^2/rho-L-in

• Prescribed or known steady wall temperatures Tw(x) < Tsat(pin) for all x-locations over which film condensation occurs.

Only the computational results described in [1]-[6] are quantitative with regard to local heat flux and film thickness variations. For experimental results to be quantitative with regard to local heat flux and associated film thickness variations, one needs to invent and integrate a non-intrusive sensor capable of measuring local time varying thickness associated with a dynamic condensate. Such a sensor has recently been invented by our research team and it is reported here. The integration of this sensor along with some other sensors in a suitable experimental facility for full quantitative investigation of shear and gravity driven internal condensing flows in a channel is part of planned research over FY06-FY09.

Recently Invented and Tested Film Thickness Sensor

The attached annual NASA report [7] presents successful design, development, and calibration of a real-time sensor for measuring a dynamic liquid film's time varying thickness using fiber-optics and fluorescence principles. FC-72 liquid, which is commonly used for electronic cooling purposes (and is being used in the above described condensing flow experiments) has been selected as a working fluid for film thickness measurements. It was doped in small concentrations by fluorescent bi-acetyl. It has been experimentally verified that, in the above experiments, the dopant adequately evaporates and condenses with the working fluid, FC-72.

Various existing film thickness measuring devices or techniques, viz. conductance probes, ultrasound sensors, reflectance based fiber-optics probes, capacitance probes, etc. were not suitable for measuring dynamic and local film thickness under the conditions of interest (e.g. wavy liquid film) and thickness ranges of interest (typically from 0.5mm to 3mm).

The report [7] describes how a liquid film can be chemically doped such that it will fluoresce in an appropriate wavelength spectrum when excited by a light source of a suitably chosen wavelength. The sensor's illuminating and receiving fiber-optic probe-tips are flush with the surface over which the liquid film of interest is present. This allows for a non-intrusive optical sensing technique.

The report [7] also presents an important paired set of calibration experiments, highlighting the developed calibration principle that is based on one-to-one correspondence among various collected optical signals and the desired film thickness measurements. This novel calibration technique does not require complete elimination of the detected noise light. However, in order for the sensor to work for wavy films, effects arising from reflections of fluorescent and noise light from the liquid-vapor interface are addressed such that these effects become nearly independent of the different interface angles that the interface is likely to experience.

The effects of the doped solution's concentration and temperature variations on the detected amount of signal and noise light have also been reported. By minimizing the errors associated with the variations in concentration and temperature, the total error of the sensed signal D can be minimized to within +/- 5.1%. This, in turn, leads to a thickness measurement whose resolution varies from +/- 0.09 mm (over 0.5 to 1.5 mm) to +/- 0.18 mm (over 1.5 to 3.0 mm). As indicated in the report [7], this resolution can be further improved.

The functionality of the invented sensor has been experimentally verified with the help of an independent, dynamic test involving a time-varying film thickness of an independently ascertained frequency and amplitude. Details and guidelines for further improvements and adaptations of this sensor are also presented in [7].

The above descriptions indicate that most of the FY05 "Task Objectives" have been met.

Bibliography Type:	Description: (Last Updated: 06/28/2013)
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